

Committee: World Health Organization

Topic: The Question of Addressing the Lasting Effects of COVID-19 and Strategies for Post-Pandemic Recovery

Student Officer: Siwoo Kim

Position: Deputy Chair of World Health Organization

Introduction

During the past few years, COVID-19, a worldwide pandemic, has left an indelible mark on the various aspects of our lives, including the economies, societies and mostly in global health. COVID-19, caused by the novel coronavirus SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2), was first identified in Wuhan, China, in December 2019. It rapidly evolved into a global pandemic, affecting millions worldwide. As of 17 December 2023, more than 772 million confirmed cases and nearly seven million deaths have been reported globally. While the immediate threat of the virus has been significantly alleviated, thanks to the widespread vaccination efforts and public health interventions, still, the lasting effect of the pandemic continues to shape national policies and individuals' lifestyles as well.

Meanwhile, the world faced several challenges, particularly in the field of overwhelmed healthcare systems, resulting in shortages of medical resources, hospital accommodations, and healthcare workforces. This striking pandemic has accelerated the adoption of innovations, including telemedicine and telehealth platforms to reduce face-to-face interactions and enhance healthcare delivery. Healthcare systems have devoted resources to invest in monitoring, testing, contact tracing, and vaccine distribution to address future outbreaks. Hospitals and facilities have modified procedures as well to handle increased cases, such as increasing beds in intensive care units and redistributing resources. The Pfizer-BioNTech vaccine, for instance, was produced in a short period, taking less than a year, exhibiting an outstanding success rate of 95% in preventing COVID-19 infections. Global vaccination campaigns began quickly, giving hope that the virus could be controlled and reduce the severity of the illness. However, global vaccine distribution revealed major inequalities. Wealthy nations obtained the majority of vaccine doses, resulting in significantly lower vaccination rates in low-income countries. The density of medical doctors

is particularly low in LEDCs. Similarly, the average density of nurses in LEDCs is less than one-third the level found in OECDs. In fact, only four LEDCs (Tuvalu, Kiribati, Lesotho and Nepal: mostly countries with small populations) are higher than the OECD average. LEDCs not only face a dearth of health professionals relative to the size of their populations but also an inadequate health systems infrastructure. Due to the relative shortage of support for the LEDCs, the recovery rates were comparatively lower than the MEDCs.

Country group	Density of medical doctors	Density of nursing and midwifery personnel	Hospital beds
Developed countries	31	113	52
Other developing countries (ODCs)	14	26	22
LDCs	3	7	6
African LDCs and Haiti	1	7	4
Asian LDCs	5	8	8
Island LDCs	4	16	11
LDCs / ODCs ratio (%)	20	28	25

Health system indicators per country group (World Health Statistics 2020 and The Global Health Observatory, July 2020)

COVID-19 also had significant health consequences, which ranged from mild symptoms to severe respiratory distress, which would even require hospitalization, and possibly death. In severe cases, it can progress to pneumonia and critical respiratory distress syndrome ARDS(Acute Respiratory Distress Syndrome), requiring intensive medical intervention. COVID-19 can also, affect organ systems, leading to various immediate health complications. There have been reports of cardiovascular problems, including myocarditis, arrhythmias, and thromboembolism, which have led to higher rates of illness and mortality among those affected by the virus. These health impacts immensely affected the vulnerable populations, such as the elderly, individuals with underlying health conditions, and frontline healthcare workers.

The WHO had submitted their plans about the recovery processes and how they will proceed with the current pandemic. Some suggestions insisted by the WHO is that they will be adding a facilitating restoration to the pre-shock state and address the pre existing gaps of the system. The WHO technical products aim to support countries to prioritize and mainstream health system recovery through effective planning as part of efforts to build health system resilience in support of universal health coverage, health security and socioeconomic development.

Today, while the global situation has drastically improved, still, COVID-19 has not disappeared. With over half a million cases confirmed every week, hundreds of thousands of people in the hospital, and thousands of people are dying each week around the world. The SARS-CoV-2 virus is highly

mutative, which in the past, new variants such as Delta and Omicron triggered increases in the number of cases and hospitalizations. However, together with the alteration of the virus, the world as well has started to enter a new phase of managing the long-term consequences of the pandemic and building resilience to prevent future health emergencies. Since the peak of the Omicron surge at the end of January 2022, daily COVID-19 reported cases have declined by over 80%. In some nations, such as the US, even after the PHE (Public Health Emergency) was brought to an end, on March 11, 2023, the government had run a general health insurance program, which distributed free vaccinations and treatments to the residents.

Definition of Key Terms

Post-Pandemic Recovery

Post pandemic recovery refers to the period after the acute infection with COVID-19, when a person, community, or even an entire society works to heal and return back to their normal function. This recovery contains diverse types of recoveries. For instance, there are Individual Health Recovery, Social and Economic Recovery, Global Recovery etc. This term would be very help for the debate, as delegates would have to find ways to revitalise the global pandemic.

Long Covid-19/Post-Covid-19 condition

Long COVID is defined as a chronic condition that occurs after SARS-CoV-2 infection and is present for at least 3 months. Long COVID includes a wide range of symptoms or conditions that may improve, worsen, or be ongoing. Post-COVID-19 conditions can affect the ability to work and may lead to loss of productivity, and a reduction in income and quality of life. It has some alterations that are quite distinct from the initial COVID pandemic. This term for sure is what should be in mind for every delegate as the whole topic is about the ways to overcome this issue, and furthermore suggests the problems that are continuously being called.

SARS-CoV-2

Stands for 'Severe acute respiratory syndrome coronavirus 2' which is the virus that causes the disease coronavirus. SARS-CoV-2 belongs to the broad family of viruses known as coronaviruses. It is a positive-sense single-stranded RNA (+ssRNA) virus, with a single linear RNA segment. This term is highly essential as the whole pandemic of COVID-19 was initiated by the emergence of the SARS-2 virus.

Frontline Healthcare Workers

They are the individuals who provide direct patient care and are the first point of contact for individuals seeking medical assistance. They work in various settings like hospitals, clinics, and emergency rooms, delivering essential medical services and support. Their major roles are direct patient care, diagnosis and treatment, preventive care, emergency response, management of chronic conditions,

referral, etc. This term as well should be something to keep in mind, as it was how people were cured back then, and could also be a possible solution.

Telehealth

It is the use of digital information and communication technologies to deliver healthcare services remotely. It allows healthcare providers to consult with patients, monitor their health, and provide education from a distance, often using technologies like video conferencing, online portals, and remote monitoring devices. Some well known platforms are Teladoc, Amwell, MDLive, Doctor on Demand, and PlushCare. Telehealth, broadly covers all remote healthcare services, including administrative, educational, and public health initiatives, in addition to clinical care. Telehealth is important as it was the technique that had provided a considerable amount of help to people during the pandemic.

Telemedicine

Telemedicine involves using technology like video conferencing, phone calls, and secure messaging to provide healthcare services. Patients can connect with their doctors, therapists, or other healthcare professionals through various digital platforms, allowing for consultations, diagnosis, treatment, and even remote monitoring. The difference between telemedicine and telehealth is that telehealth not only includes telemedicine, but also includes non-clinical services, when Telemedicine specifically refers to remote clinical services, like diagnosis, treatment, and monitoring, delivered by licensed healthcare professionals.

PHE(Public Health Emergency)

An event or situation that poses a significant risk to the health of a large number of people, potentially causing illness, injury, or death. These events can be natural, like disease outbreaks or natural disasters, or man-made, such as bioterrorism or chemical attacks. Declarations of public health emergencies often trigger special legal and regulatory responses to mitigate the impact of the event. Pandemics like COVID-19, seasonal influenza, and outbreaks of infectious diseases like SARS or Ebola are some examples when the PHE was declared.

Omicron

A variant of SARS-CoV-2, the virus that causes COVID-19. It first appeared in November 2021 and quickly became the dominant variant around the world. It spreads faster than previous COVID-19 variants but causes less severe illness. Its symptoms are similar to those of other variants of this virus. The name of these bacterias are some of the mutations of SARS-CoV-2 virus which provoked sequential problems.

Delta

The Delta variant was a variant of SARS-CoV-2, the virus that causes COVID-19. It first appeared in India on 5 October 2020. Unlike the Omicron, Delta caused more severe disease than other variants in people who weren't vaccinated

Contact Tracing

It is the process of quickly identifying, assessing, and managing people who have been exposed to a disease to prevent additional transmission. During the process of the contact tracing, once a person tests positive for a disease, they will be asked to list the people they have been in contact with and the places they visited during the period in which they were contagious. This term suggests how people during the pandemic had tracked the virus. It as well could be claimed as a possible solution for this topic.

Myocarditis

Myocarditis is a serious condition where inflammation develops in the myocardium, the middle muscular layer of the heart wall. As a result, the heart's ability to pump blood is reduced. The condition may be acute and resolve quickly , or it may be subacute or chronic, lasting up to three months. In severe cases, myocarditis may lead to heart failure or death. Those with mild myocarditis may need only rest and medication. People with severe cases may need lifelong medication, implanted devices or other surgeries. Medical terms as well are important as these were the major types of diseases people had suffered from during the pandemic.

Arrhythmia

An arrhythmia is a problem with the rate or rhythm of your heartbeat. Your heart may beat too quickly, too slowly, or with an irregular rhythm. Arrhythmias are treatable with medicine or

procedures to control the irregular rhythms. If not treated, arrhythmias can damage the heart, brain, or other organs. This can lead to life-threatening stroke, heart failure, or cardiac arrest.

Thromboembolism

It is a condition where a blood clot forms, breaks loose, and travels through the bloodstream to obstruct a blood vessel, causing a blockage. This blockage can disrupt blood flow to vital organs, potentially leading to tissue damage or even death. There are 2 types of thromboembolism, Deep Vein Thrombosis and Pulmonary Embolism. Deep Vein Thrombosis forms in a deep vein, usually in the legs or pelvis. Pulmonary Embolism clot travels to the lungs and blocks a pulmonary artery, which is a serious and potentially life-threatening condition.

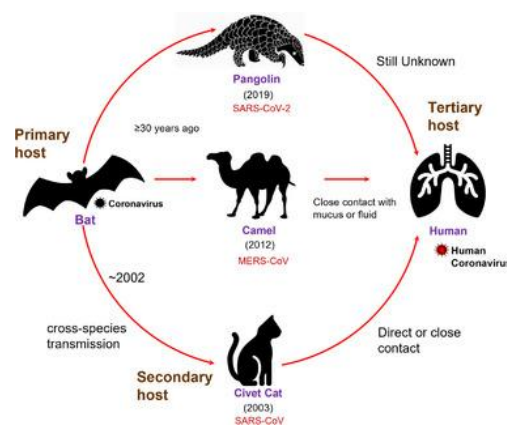
Pre-Shock State

The earliest phase of shock where the body's compensatory mechanisms are still working to maintain adequate blood flow and oxygen delivery to tissues despite an initial decrease in blood volume or pressure

Background Information

SARS-CoV-2 Virus

SARS-CoV-2 (Severe Acute Respiratory Syndrome coronavirus 2) is the virus that causes the disease COVID-19. In fact, SARS-CoV-2 is not only the one that causes coronavirus, but also is one of the many different types of COVID. SARS-CoV emerged at the end of 2019 in Wuhan, China. Back then, it was not precisely proven how the virus had first appeared, and how humans had first come in contact with it. However, with all the evidence provided, they were passed from bats to other animals, which eventually came in contact with humans. As viruses easily and rapidly adapt to their new surroundings, it is likely that the viruses have adapted to the environment of humans, to transmit it from person to person. People who are infected with SARS-CoV-2 release virus particles or germs through their mouth, especially their saliva or nose, through coughing, sneezing, or talking.



Schematic diagram of the transmission process of the virus (National Academy of Sciences of the United States of America, 2022)

Long-term health effects

The term “Long-term COVID” in this case refers to the unsolved issues within the conditions, where each individual experiences continuous symptoms that have persisted for an extended period of time. COVID-19, caused by the perilous virus, has affected billions of people worldwide, not only during the breakout but even in the following years. While most global attention was focused on the seizure of the spread of virus, the issue of long-term health consequences, as another critical issue, had emerged. Usual immediate health issues the victims had faced ranged from mild symptoms to severe respiratory

distress or death. Many reports have underscored that the capacity of the virus can also affect the central nervous system. However, the symptoms and consequences between the initial effect and the long-term effect are significantly different. For the long term health effect, the virus of COVID-19 had sequentially caused a course of residual lung damage such as lung fibrosis (a lung disease caused by the accumulation of fibers which are connected to the tissue of the organ, eventually causing permanent scarring or an organ malfunction), bronchiectasis(a condition where damage causes the tubes in lungs to widen or develop pouches, which makes it hard to clear mucus out of the lungs and can cause frequent infections.), or other structural abnormalities.

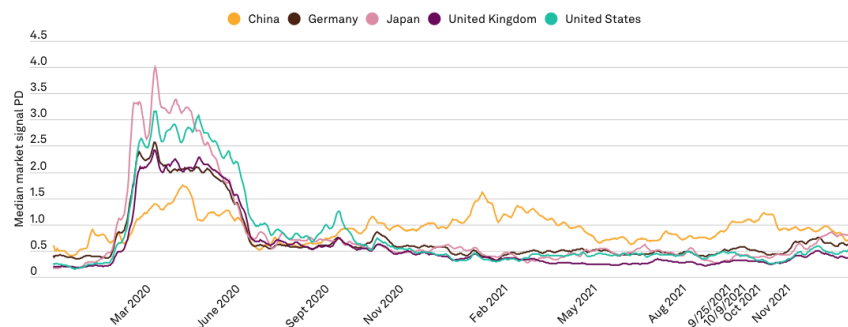
Not only that, they could notice that the long-term COVID effect was very alarming because they are so unpredictable. In fact, there were some astonishing reports that were reported about the COVID cases. COVID actually didn't affect people who were hospitalized or had severe symptoms, but affected people who were very young, healthy, and experienced only mild symptoms. This was because of the immune system responses humans have made towards the virus. By fighting off the powerful diseases, the human body was able to balance its immune system to the harsh environment.

There were not only physical changes, but also a significant effect on an individual's mental health. There were 2 kinds of cases for the consequence of the mental health outcome: the virus's infection to the brain and the PTSD(post-traumatic stress disorder) due to the large-scale pandemic. First, there was a possibility that the virus may have infected the brain cells or triggered an immune response that caused an adverse effect on the brain function and mental health. Research from 'Brain & Behavior Magazine' has suggested that there was a COVID-19 protein found in the linings of blood vessels in the brain of an individual. These penetrations of viruses to our brain may cause diseases such as neuroinflammation or brain fog, which can damage one's mental health to be damaged.

Second, is the case of an indirect effect, PTSD. People who were facing mental problems were mostly struggling with depression or anxiety. Moreover, patients who were hospitalized for SARS were insomnia(41.9%), followed by impaired concentration or attention(38.2%), anxiety(35.7%), impaired memory(34.1%) and depressed mood(32.6%). Throughout the world, such symptoms mentioned previously were similarly found. In Hong Kong as an example, compared to the baseline of 3% having psychiatric morbidities, a marked increase in the prevalence of mental disorders at 42.5% was found at around 4 years after the phase. Sequentially, PTSD, depressive disorders, anxiety disorder, panic disorder, and somatoform pain disorder were followed, with the ranges from 15%~30%.

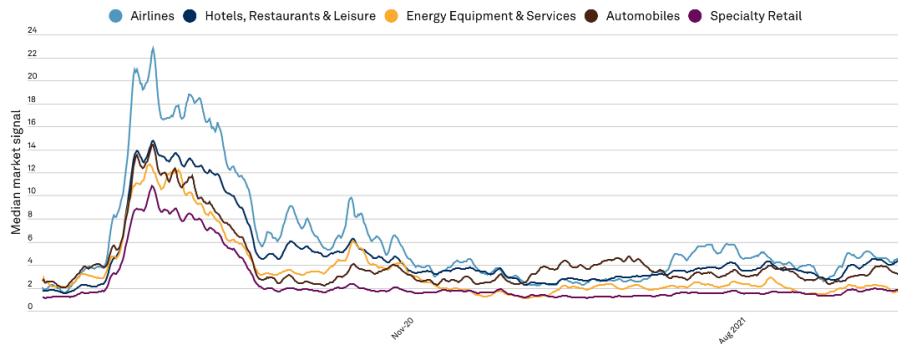
The Aftermath in the Economy

The COVID-19 pandemic has brought up profound and destructive effects to the economy worldwide. The IMF(International Monetary Fund) for instance has documented a significant decline of approximately 3.5% in the worldwide GDP for the year of 2020. This number, in fact is showing that, global GDP reached 84.9 trillion U.S. dollars in 2020, meaning that a 3.4 percent drop in economic growth results in over two trillion U.S. dollars of lost economic output. The economic recession had a significant impact on several nations, resulting in a 4.3% contraction for the United States' GDP. The global stock markets as well have suffered dramatic falls after the coronavirus outbreak, although they were able to recover from the losses quite quickly. The Dow Jones reported its largest ever single-day loss of almost 3,000 points on March 16, 2020, which beats its previous record of 2,300 points that was set only four days earlier. Not only the US but also in regions like India and the UK, had experienced an 8% economic contraction.



Countries Overview of COVID-19 Impacts (S&P Global Market Intelligence, Jan 15 2022)

The COVID-19 pandemic had a varied impact on different sectors and industries. Usually, one of the best and most effective ways for most nations to earn money is by tourists who tend to visit the nation. As more people visit, nations could gain more profits. However, as countries around the world closed their borders and imposed travel restrictions, especially the travel and tourism industry was heavily affected. The travel restrictions led to a sharp decrease in the number of flights worldwide. To be more specific, the governments did not allow citizens to freely travel around unless they were proved to be unaffected. This basically means that the travel agency and the nation would not have the chance to earn money and increase their GDP.



Top 5 industries that were most affected by the COVID-19 pandemic (S&P Global Market Intelligence, Jan 15 2022)

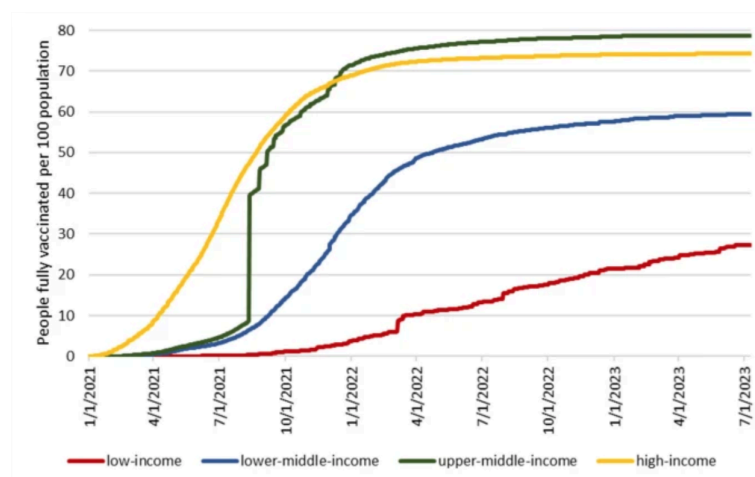
While the pandemic had hit the whole global economy, some countries and regions were either less affected or had a positive GDP growth rate. For instance, in the case of China, the origin of this whole pandemic had gained nearly 4.3% of GDP growth rate during the 2020s which tended to grow more to nearly 8% during the year of 2021; the increase of GDP was supported due to their exports, especially labor-intensive products or medical supplies as an engine of economic growth. In Asia, the change in the GDP varied between 0.2 percent in East Asia to minus 7.7 percent in South Asia. The reasons behind this are that the East Asian countries' relatively fast response to the pandemic in its early stages meant that they could ease restrictions quite fast. However, China chose to hold on to its zero-COVID policy, meaning that the country's economy was unable to reopen fully when the global economy did so. This led to slowed growth to about 3% which eventually declined by almost two percent in the final quarter of 2022, compared to the same quarter in 2021.

Inequality in vaccine access

One of the other challenges nations faced during the COVID-19 pandemic was the unequal distribution of vaccines, which highlighted deep global and domestic disparities. The GHG (Global Health Governance) is responsible for the equitable accommodation of COVID-19 vaccine distribution. However, this was not what was shown from the case. According to the Our World in Data website on April 7 2022, the share of vaccination among the people in high-income countries and upper middle income countries has reached 74.1% and 76.68%. On the other hand, the share of vaccinations among people in lower middle income countries and low-income countries reached only 50.51% and 11.51% respectively, demonstrating the drastic difference in the accessibility of vaccines. Several studies have published the reasons underlying the inequity in COVID-19 distribution.

Some studies have discussed the influence of economic power in not only securing the vaccine but also controlling its distribution. Nations with high economic power can produce, trade, and control the value chain of vaccines. The financing capacity enables the production as well as the purchase of any required vaccines. Countries' purchasing power appeared to be determinative in the availability of the vaccines. Not only that, but even the political power imbalances the accessibility. The powerful nations would compete to settle their ideological perspectives in the arena of global health and use their power to influence the less powerful nations' access to vaccines. Therefore, countries with higher vaccine coverage are expected to have a better chance for rapid economic recovery, which are the nations with strong economic powers.

On another aspect, there were some studies that have published the reasons for the underlying inequalities in COVID-19 vaccine distribution, mostly related to the countries' health systems. Their systems mostly highlighted the countries' limited capacities, poor infrastructure, inadequate supply chain capability and limited technical expertise as the main barriers to the storage, manufacturing and administration of the COVID-19 vaccine usage. Therefore, the low income and lower middle income countries have relied mostly on vaccine donations from developed countries and on support from COVAX (COVID-19 Vaccines Global Access), a global vaccine sharing scheme led by the Coalition for Epidemic Preparedness Innovations (CEPI). These inequities in the distribution of the cure had led to a significant record of death as people were not getting treated right away due to all the delays they had faced.



Percentage of fully vaccinated individuals by income group(Our World in Data, 2021)

Possible solutions

The severity of the COVID pandemic has revealed some systematic vulnerabilities in the global health preparedness and highlighted the inequalities in access to medical resources. In order to mitigate the lasting effects and ensure resilience against future health crises, nations should consider diverse solutions.

Health System resilience

In the most common form, the word 'resilience' refers to the capacity to recover quickly from a shock, and until now, health system resilience has typically focused on understanding the health system's preparedness. However, this new field of study, about the COVID-19 pandemic, is focused on strengthening the health systems. In order to establish the health system resilience, there should be an enhancement in the leadership and governance of the COVID-19 responses. Strategies to achieve effective implementation of public health measures as well as the FTTIS(Find, Test, Trace, Isolate and Support) systems, or even the income support, should be developed which cover the strategies to maintain regular health services, such as childhood vaccination programs and screening services. Radical policy denials have been a feature of the COVID-19 pandemic in some countries, often happening as a delayed response to pressures from the public. Transparent communication throughout the response, particularly about the uncertainties underlying policy decisions, may help solicit public support for adaptations to response strategies.

International Cooperation

Such major issues as COVID-19 cannot be resolved by the efforts of a single nation. Nations and parties would have to facilitate international cooperation to resolve the global crisis. At the center of the WHO, its networks of researchers and other experts have coordinated efforts on surveillance, epidemiology, modeling, diagnostics, clinical care and treatment, and on researching other ways to identify and manage the disease and limit its transmissions. This has thereby reduced the probability of the high lethality of viruses such as SARS-CoV-2 virus. In particular, as the four leading national scientific funding organizations or programs related to the health domain in the world, the NIH(National Institutes of Health), the framework programs for research and innovation, funded by the EU, and the KAKENHI (Grants-in-Aid for Scientific Research)program of the Ministry of Education, Science and

Culture of the Japanese government are significant, there are many words and ideas being gathered about how these organisations should provide funding making international cooperations. For instance, digital health technologies, such as electronic health records and remote monitoring devices, could be adopted to enhance patient care and efficiency.

More Investment in Health Care Infrastructure

From the aftermath of the COVID-19 pandemic, it was globally exposed that there were severe systemic weaknesses in the aspects of infrastructure, supply chains, government preparedness and actions, human resources, and public health systems. Among other setbacks in the outbreak of the pandemic, it became significantly clear that the health care facilities were unqualified and unprepared to manage the influx of patients. Therefore, part of the recovery from the pandemic required restructuring the infrastructure. With the support of infrastructure investment, it would not only be robust but also be innovative and adaptable to the changing circumstances and complexities. With the enhancement of infrastructures, for instance, more workspace for medical volunteers to support without limits, it would alleviate the seriousness of the aftermath with a more fast and efficient treatment. As the infrastructures develop, there will be more training systems established for the healthcare workers, leading to an expanded range of services.

This webinar is undertaken as part of the Response and Recovery: Mobilising Financial Resources for development in the time of COVID-19 in cooperation with ECA (Economic Cooperation Administration), ECLAC (Economic Commission for Latin America and the Caribbean) and ESCAP (The Economic and Social Commission for Asia and the Pacific), aims to strengthen macroeconomic diagnostic and policy design capacity in low-income and middle-income developing countries (LICs and MICs) to respond appropriately and innovatively to the COVID-19 pandemic and to contribute to a recovery that enables the achievement of the Sustainable Development Goals (SDGs).

More Transparent Vaccine Usage

Treatments that provide an essential recovery factor are necessary for such pandemics, especially the current pandemic, COVID-19. However, in such LEDC nations, where exports and transfer of vaccinations are limited, the majority of the citizens face issues in getting treated properly.

In order to resolve these issues, strong leadership engagement and commitment to detailed and cost vaccination plans are essential. For instance nations such as Canada which had received numerous vaccines could provide help to other nations. The COVID-19 vaccination services will need to be

integrated with other immunization services and alongside other health and social interventions for maximum impact and to build long-term capacity.

Increased Education Effort

Providing students or teenagers with the right education style based on the current issues are important as well. For instance, providing an early childhood program during the elementary, secondary school, and postsecondary institutions, children would be able to better adapt to the drastic changes that have been made and changes that would be made in the future. In fact, the enclosure of schools was the most significant interruption to children, which resulted in children's increased anxiety, reduced social interactions and peer relationships, lost learning opportunities in the school environment, and a widening of achievement gaps.

In order to overcome these drawbacks, schools should widen the education opportunities for children program enrollment. For instance, the education department could enhance the online education program, for all students who are missing out their classes, or even add a subject or class that educates them about the current pandemic. This would help the students acknowledge the current situation and not get confused when they are set in a certain restriction.

Major Parties Involved

People's Republic of China:

China is known as the initial nation that has caused the spread of the pandemic as on December 29th 2019, a cluster of infected pneumonia cases were reported in Wuhan. Within a first few weeks, China had experienced an exponential development in the infection with a peak of 3893 new cases reported. This eventually led China to be one of the top ranking nations with recorded cases. However, as the nation was well prepared and implemented a quick lockdown, keeping the cases below 100 per day has been available, therefore leading to maintaining their nation's well-preserved aspect.

China initially had consisted of hospitals, clinics, and public health institutes as of 2003, since the outbreak of the disease SARS (severe acute respiratory syndrome). Nowadays, China has developed its infrastructure and services, leading to a larger-scale medical system, providing 79% of sickbeds in the country.

United States of America

The first COVID-19 case in the US was reported in Washington state on January 20, the same day as the first reported case in South Korea. Twelve weeks later, on April 11, the US surpassed Italy as the country with the most reported COVID-19 deaths (approximately 24,000,

In addition to testing, the US has also fast-tracked vaccine development, spurring high-ranking officials such as the country's leading public health official, Dr. Anthony Fauci, to say that a vaccine may be available for the US population as soon as January 2021. To do so the government introduced 'Operation Warp Speed', which will fund the production of vaccines while they are still being tested for efficacy and safety, trading off the speed at which the vaccine will be disseminated against a wasteful investment in all the vaccine candidates that do not pass efficacy and safety standards.

Mitigation efforts such as stay-at-home orders and non-essential business shutdowns severely impacted the US economy and financial markets. For instance, the first quarter of 2020 saw a 4.8 percent decrease in US GDP, and the second quarter saw an even more dramatic decline of nearly 33 percent.

United Kingdom

The number of daily cases exceeded three figures on 21st March, which peaked on 1st May. UK-wide COVID-19 cases were 313,798 by 12th August, and overall deaths were 46,706. The proportion of deaths-to-cases for the UK up to 12th August was 15%. COVID-19 cases subsequently increased exponentially, and a host of measures have been rapidly implemented to decelerate the spread of SARS-CoV-2 to reduce the impact on the healthcare system, including social distancing, self-isolation and lockdown strategies. Financial interventions were rapidly introduced to protect the UK economy.

The Government has said that the vaccines pledged would be drawn from the UK's stocks and expected deliveries of various vaccines. To be more specific, the UK has donated 5 million doses by the end of September, primarily for use in the world's poorest countries. The Prime Minister has also committed to donating a further 95 million doses within the next year, including 25 million more by the end of 2021. 80% of the 100 million doses was gone to Covax and the remainder will be shared bilaterally with countries in need.

Republic of the Sudan

The first case that was reported by Sudan was on 13 March 2020, and up to 3 July 2020 there were 9894 confirmed cases and 616 deaths. There is variation in case of fatality rate, which in some cities was low (3.8%), but in others it was very high (31.7%). The government of Sudan has implemented preventive measures during the current coronavirus disease pandemic, such as partial lockdown, contact monitoring, risk communication, social distancing, quarantine and isolation to prevent the spread of SARS-CoV-2. However, there are new community cases every day; this could be as a result of the weak application of these measures by the government, and the lack of commitment of people to these measures

Comparing the percentage of positivity in Sudan with other countries such as Korea (1%), Australia (1%), Mexico (20%) and Bolivia (50%), we find that Sudan is among the countries with the highest positive rates. This high positivity rate (55%) indicates that Sudan is not testing widely enough to find all cases, and the actual number of circulating cases in the community is far higher than the confirmed numbers

Moreover, UNICEF have reported that Sudan has received 3.4 million doses of the AstraZeneca vaccine, meaning that they have barely gotten a quarter of the immunizations it requires. To be more specific, the Sudanese government has vaccinated around 830,000 individuals out of a total population of 45 million since March.

Dominion of Canada

Canada during the pandemic is ranked as one of the best nations that have dealt with the issue. The Canadian health infrastructures offered on average 12.9 intensive care unit beds per 100,000. In 2020, Canada reported 1140 hospitals across the country, with 124.5 family medicine physicians per 100,000. Moreover, due to the novelty of the virus, research was quickly organized to find the causes and solutions to the spread of the virus.

Canada as well, has implemented strict border controls, mandatory quarantine measures for travelers, and large-scale public precautions. Although the nation has partially faced a significant shortage in equipment, the federal and provincial governments have coordinated responses to strengthen the healthcare systems. By mid-2021, Canada was ranked to have one of the highest vaccination rates in the world, with the majority of its population receiving at least one dose within months of the vaccines becoming available.

Timeline Of Events

Date	Description of event
<i>Wuhan, Hubei Province Dec, 12, 2019</i>	A crowd of patients in China's Hubei Province, which is a city of Wuhan began to experience symptoms of a strange pneumonia-like illness, experiencing shortness of breath and fever, which were not responding well to the original standard treatments.
<i>Declaration of SARS-CoV2 virus Jan 31, 2020</i>	U.S. Secretary of Health and Human Services declares the SARS-CoV2 virus a public health emergency and the 'White House 2019 Novel Coronavirus Task Force' announces implementation of new travel policies effective February 2
<i>CDC sample kit revision Feb, 9, 2020</i>	After the first sample CDC kit has been refused due to the untrustworthy result, an evaluated version and guidance has been updates, to prepare for any additional spread of the virus
<i>Announcement of the vaccine trials, July, 27, 2020</i>	Pfizer and Moderna each announced the start of their 3 clinical trials for the Pfizer COVID-19 vaccine BNT162b2 and Moderna mRNA-1273 vaccine.
<i>Criteria announced by the CDC, May 13, 2021</i>	CDC (Centers for disease control and prevention) announced that those fully vaccinated can take off their masks indoors and outdoors when gathering of any size.
<i>New variant of the virus announced, June 1, 2021</i>	The Delta variant, which is the mutant of the SARS-CoV-2 virus, became the dominant variant in the US and kicked off another wave of infection.
<i>The positive outcomes of the vaccines, June 7, 2021</i>	CDC (Centers for disease control and prevention) finds that the Pfizer-BioNTech and Moderna mRNA COVID-19 vaccines reduce the risk of infection with the SARS-CoV-2 virus by 91% and protect against severe illness and hospitalization if a

	breakthrough infection does occur.
<i>Announcement for the age restrictions on the usage of vaccines., August 23, 2021</i>	The FDA (Food and Drug Administration) fully approves the Pfizer-BioNTech COVID-19 vaccine for all people ages 18 years and older. Full FDA approval further reinforces that the Pfizer-BioNTech COVID-19 vaccine has been shown to meet the agency's high standards for safety, effectiveness, and consistent quality in manufacturing.
<i>Announcement of the modified age restrictions, September ~ November, 2021</i>	People who are in the ranges of 5~11 years old and 65years or older were fully approved to get vaccinated by the Pfizer-BioNTech.
<i>US requirement announced for travelers, Nov 8, 2021</i>	All non citizens who are travelling to the US were then required to be fully vaccinated and provide proof of their vaccination status to fly to the US. All travelers were continuously required to show a negative pre-departure COVID-19 test taken no more than three days before they board their flights.
<i>New variant of SARS-CoV2 virus announced by WHO, Nov 26, 2021</i>	WHO designates the COVID-19 "Omicron" variant, first identified by scientists in South Africa, as a "variant of concern." Changes in the spike protein of the Omicron variant of the SARS-CoV-2 virus, concern scientists around the world due to the potential for increased transmissibility and decreased vaccine protection.
<i>CDC announces the quarantine period for workers, December 23, 2021</i>	CDC updates its recommendations for quarantine periods for healthcare workers, decreasing their isolation time to 7 days with a negative test and healthcare workers who have received all recommended COVID-19 vaccines doses, including a booster, do not need to quarantine after a high-risk exposure.
<i>Study of the symptoms of the disease gets published, February 7 2022</i>	A study is published in Nature showing that even a mild case of COVID-19 appears to increase the risk of heart problems for one year after infection. The study's authors suggest that COVID-19 might be as much of a risk factor for heart disease as high blood pressure, diabetes, or smoking.

<i>Announcement of WHO about the inequity in vaccine distribution, March 5, 2022</i>	More than 10 billion people have received a COVID-19 vaccine, with WHO reporting that 10,704,043,684 COVID-19 vaccine doses have been administered worldwide. About 56% of the world is now fully vaccinated, but many regions still lack access, especially on the African continent where less than 20% of the total population is currently vaccinated.
<i>China's reestablishment after the lockdown , May 31, 2022</i>	Authorities in Shanghai announce that they are partially reopening China's largest city after two months of a COVID-19 lockdown that has kept millions of people strictly in their homes and employed both mass testing and isolation of anyone infected with COVID-19 in centralized facilities.

UN Involvement, Resolutions, Treaties and Events

- Draft Resolution COVID-19 response, 18 May 2020, A73/CONF./1 Rev.1

The draft resolution submitted by the World Health Organization suggests the need for international solidarity and cooperation in combating the pandemic. The resolution calls for equitable access to essential supplies, diagnostics, and vaccines. It requests the WHO to continue with the leading and coordination of global efforts, as well as supporting member state;s responses. With more soared information, protected healthcare workers and menstruation of transparency, the issue about the inequalities would be alleviated and ensure preparedness for future crises.

- Draft Resolution Global solidarity to fight the coronavirus disease 2019 (COVID-19), 3 April 2020, A/RES/74/270

The draft resolutions have emphasized the need for global solidarity, multilateral cooperation, and the unwavering support for human rights in the face of the cOVID 19 pandemic. It condemns all forms of discrimination, including racism and prejudice, and calls for a united international response to the crisis. It also insists that the Member States should coordinate actions, share vital information, and work together unitedly to address the various impacts of the pandemic.

- Draft Resolution United Nations Security Council, 26 February 2021, S/RES/2565 (2021)

The draft resolution urges all parties in conflict to implement a humanitarian pause, allowing safe and unhindered delivery of vaccines. It proclaims that widespread vaccination should be considered a global public good essential for terminating the pandemic. The council also underscores the importance of strengthening international cooperation, particularly supporting the COVAX Facility, to expand vaccine access

- The HLPF reviewed progress on five SDGs. The consideration of SDGs 4 (quality education), 5 (gender equality), 14 (life below water), 15 (life on land), and 17 (partnerships for the Goals) will consider the different impacts of the COVID-19 pandemic across all SDGs and the integrated interlinked nature of goals.

Bibliography

COVID019 pandemic data collection, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7451131/>.

“CDC Museum COVID-19 Timeline | David J. Sencer CDC Museum.” *CDC*,
<https://www.cdc.gov/museum/timeline/covid19.html>. Accessed 20 July 2025.

“COVID-19 pandemic in China: Context, experience and lessons.” *PubMed Central*,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7451009/>. Accessed 29 August 2025.

“Global and U.S. Agencies Declare End of COVID-19 Emergency.” *Pfizer*, 11 May 2023,
<https://www.pfizer.com/news/announcements/global-and-us-agencies-declare-end-covid-19-emergency>. Accessed 26 August 2025.

“Myocarditis.” *American Heart Association*, 24 September 2024,
<https://www.heart.org/en/health-topics/myocarditis>. Accessed 26 August 2025.

“Post COVID-19 condition (long COVID).” *World Health Organization (WHO)*, 26 February 2025,
[https://www.who.int/news-room/fact-sheets/detail/post-covid-19-condition-\(long-covid\)](https://www.who.int/news-room/fact-sheets/detail/post-covid-19-condition-(long-covid)).
Accessed 4 July 2025.

“Post COVID-19 condition (long COVID).” *World Health Organization (WHO)*, 26 February 2025,
[https://www.who.int/news-room/fact-sheets/detail/post-covid-19-condition-\(long-covid\)](https://www.who.int/news-room/fact-sheets/detail/post-covid-19-condition-(long-covid)).
Accessed 26 August 2025.

“Post-COVID pandemic: The new normal and aftermath.” *PubMed Central*,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11610867/>. Accessed 20 July 2025.

“Resolution adopted by the General Assembly on 2 April 2020.” <https://docs.un.org/en/A/RES/74/270>.

Sanchez, Jose. “Impact of the coronavirus pandemic on the global economy - Statistics & Facts.” *Statista*,
30 May 2025,
<https://www.statista.com/topics/6139/covid-19-impact-on-the-global-economy/#topicOverview>.
Accessed 26 August 2025.

“SARS-CoV-2 and COVID-19 | BCM.” *Baylor College of Medicine*,
<https://www.bcm.edu/departments/molecular-virology-and-microbiology/emerging-infections-and-biodefense/specific-agents/sars-cov-2-and-covid-19>. Accessed 20 July 2025.

“SARS-CoV-2 and COVID-19 | BCM.” *Baylor College of Medicine*,
<https://www.bcm.edu/departments/molecular-virology-and-microbiology/emerging-infections-and-biodefense/specific-agents/sars-cov-2-and-covid-19>. Accessed 26 August 2025.

“Short- and potential long-term adverse health outcomes of COVID-19: a rapid review.” *PubMed Central*, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7586446/>. Accessed 26 August 2025.

“Statement on the fifteenth meeting of the IHR (2005) Emergency Committee on the COVID-19 pandemic.” *World Health Organization (WHO)*, 5 May 2023,

[https://www.who.int/news/item/05-05-2023-statement-on-the-fifteenth-meeting-of-the-international-health-regulations-\(2005\)-emergency-committee-regarding-the-coronavirus-disease-\(covid-19\)-pandemic](https://www.who.int/news/item/05-05-2023-statement-on-the-fifteenth-meeting-of-the-international-health-regulations-(2005)-emergency-committee-regarding-the-coronavirus-disease-(covid-19)-pandemic). Accessed 26 August 2025.

Vidovic, Luka. "Industries Most and Least Impacted by COVID-19 from a Probability of Default Perspective - January 2022 Update." *S&P Global*,
<https://www.spglobal.com/market-intelligence/en/news-insights/research/industries-most-and-least-impacted-by-covid-19-from-a-probability-of-default-perspective-january-2022-update>.
Accessed 26 August 2025.